



Disciplina: SIMULAÇÃO E MÉTODOS ANALÍTICOS

Unidade de Aprendizagem: UA2 | FNDAMENTOS DE SIMULAÇÃO POR COMPUTADOR

Módulo de Aprendizagem: M2 | GERAÇÃO DE NÚMEROS PSEUDOALEATÓRIOS

Estudantes:

1. **Felipe Freitas Silva**

2.

3.

4.



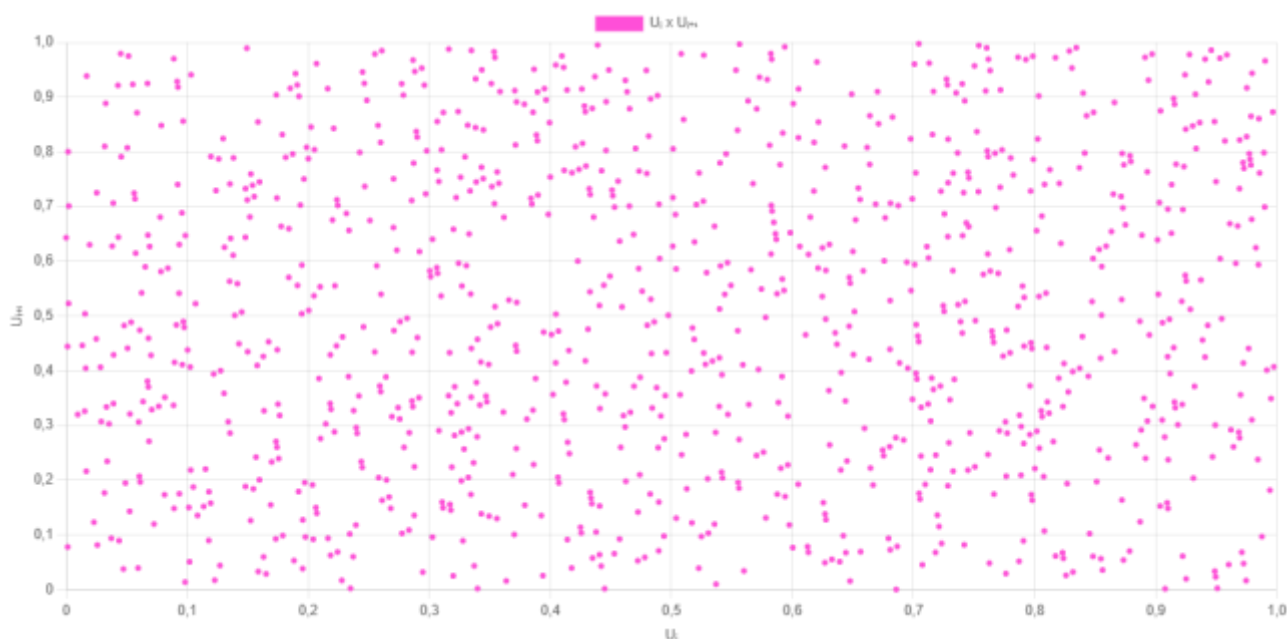
Importante!

A equipe deve registrar o nome dos estudantes que efetivamente participaram da elaboração da atividade avaliativa. Atividades encaminhadas sem a identificação dos autores não serão avaliadas. Os estudantes sem efetiva participação terão a pontuação zerada.

Atividade prática | Desenvolvendo um gerador de números pseudoaleatórios

Registre neste espaço sua resposta! ▼

1. Gráfico de dispersão dos números





2. Valores utilizados para as variáveis X_0 (semente), a , c , e M

$$X_0 = 42$$

$$a = 39758$$

$$c = 58739$$

$$M = 1000000$$

3. Os 1.000 números pseudoaleatórios uniformizados entre 0 e 1 produzidos pelo seu método

i, U_i, X_i

1,0.728575,728575

2,0.743589,743589

3,0.670201,670201

4,0.910097,910097

5,0.695265,695265

6,0.404609,404609

7,0.503361,503361

8,0.685377,685377

9,0.277505,277505

10,0.102529,102529

11,0.406721,406721

12,0.472257,472257

13,0.052545,52545

14,0.142849,142849

15,0.449281,449281

16,0.572737,572737

17,0.936385,936385

18,0.853569,853569

19,0.255041,255041

20,0.978817,978817

21,0.865025,865025



22,0.722689,722689

23,0.728001,728001

24,0.922497,922497

25,0.694465,694465

26,0.598209,598209

27,0.652161,652161

28,0.675777,675777

29,0.600705,600705

30,0.888129,888129

31,0.291521,291521

32,0.350657,350657

33,0.479745,479745

34,0.760449,760449

35,0.990081,990081

36,0.699137,699137

37,0.347585,347585

38,0.343169,343169

39,0.771841,771841

40,0.913217,913217

41,0.740225,740225

42,0.924289,924289

43,0.940801,940801

44,0.424897,424897

45,0.113665,113665

46,0.151809,151809

47,0.680961,680961

48,0.706177,706177

49,0.243905,243905

50,0.233729,233729

51,0.656321,656321

52,0.069057,69057

53,0.626945,626945



54,0.138049,138049

55,0.610881,610881

56,0.465537,465537

57,0.878785,878785

58,0.792769,792769

59,0.968641,968641

60,0.287617,287617

61,0.135425,135425

62,0.285889,285889

63,0.433601,433601

64,0.167297,167297

65,0.452865,452865

66,0.065409,65409

67,0.589761,589761

68,0.776577,776577

69,0.207105,207105

70,0.139329,139329

71,0.501121,501121

72,0.627457,627457

73,0.494145,494145

74,0.275649,275649

75,0.311681,311681

76,0.871937,871937

77,0.529985,529985

78,0.202369,202369

79,0.845441,845441

80,0.102017,102017

81,0.050625,50625

82,0.807489,807489

83,0.206401,206401

84,0.149697,149697

85,0.712065,712065



86,0.339009,339009

87,0.378561,378561

88,0.886977,886977

89,0.490305,490305

90,0.604929,604929

91,0.825921,825921

92,0.025857,25857

93,0.081345,81345

94,0.173249,173249

95,0.092481,92481

96,0.918337,918337

97,0.301185,301185

98,0.571969,571969

99,0.402241,402241

100,0.356417,356417

101,0.485825,485825

102,0.489089,489089

103,0.259201,259201

104,0.372097,372097

105,0.891265,891265

106,0.972609,972609

107,0.047361,47361

108,0.037377,37377

109,0.093505,93505

110,0.630529,630529

111,0.630721,630721

112,0.264257,264257

113,0.388545,388545

114,0.830849,830849

115,0.953281,953281

116,0.604737,604737

117,0.192385,192385



118,0.901569,901569

119,0.639041,639041

120,0.050817,50817

121,0.441025,441025

122,0.330689,330689

123,0.592001,592001

124,0.834497,834497

125,0.990465,990465

126,0.966209,966209

127,0.596161,596161

128,0.227777,227777

129,0.016705,16705

130,0.216129,216129

131,0.915521,915521

132,0.342657,342657

133,0.415745,415745

134,0.248449,248449

135,0.894081,894081

136,0.931137,931137

137,0.203585,203585

138,0.191169,191169

139,0.555841,555841

140,0.185217,185217

141,0.916225,916225

142,0.332289,332289

143,0.204801,204801

144,0.536897,536897

145,0.009665,9665

146,0.319809,319809

147,0.024961,24961

148,0.458177,458177

149,0.259905,259905



150,0.361729,361729

151,0.680321,680321

152,0.261057,261057

153,0.162945,162945

154,0.426049,426049

155,0.914881,914881

156,0.897537,897537

157,0.334785,334785

158,0.440769,440769

159,0.152641,152641

160,0.759617,759617

161,0.911425,911425

162,0.493889,493889

163,0.097601,97601

164,0.479297,479297

165,0.948865,948865

166,0.033409,33409

167,0.333761,333761

168,0.728577,728577

169,0.823105,823105

170,0.067329,67329

171,0.925121,925121

172,0.019457,19457

173,0.630145,630145

174,0.363649,363649

175,0.015681,15681

176,0.503937,503937

177,0.585985,585985

178,0.650369,650369

179,0.429441,429441

180,0.774017,774017

181,0.426625,426625



182,0.815489,815489

183,0.270401,270401

184,0.661697,661697

185,0.808065,808065

186,0.107009,107009

187,0.522561,522561

188,0.038977,38977

189,0.706305,706305

190,0.332929,332929

191,0.649921,649921

192,0.617857,617857

193,0.817345,817345

194,0.061249,61249

195,0.196481,196481

196,0.750337,750337

197,0.957185,957185

198,0.819969,819969

199,0.386241,386241

200,0.228417,228417

201,0.461825,461825

202,0.297089,297089

203,0.723201,723201

204,0.084097,84097

205,0.587265,587265

206,0.540609,540609

207,0.591361,591361

208,0.389377,389377

209,0.909505,909505

210,0.158529,158529

211,0.854721,854721

212,0.056257,56257

213,0.724545,724545



214,0.518849,518849

215,0.457281,457281

216,0.636737,636737

217,0.448385,448385

218,0.949569,949569

219,0.023041,23041

220,0.122817,122817

221,0.017025,17025

222,0.938689,938689

223,0.456001,456001

224,0.746497,746497

225,0.286465,286465

226,0.334209,334209

227,0.540161,540161

228,0.779777,779777

229,0.432705,432705

230,0.544129,544129

231,0.539521,539521

232,0.334657,334657

233,0.351745,351745

234,0.736449,736449

235,0.798081,798081

236,0.163137,163137

237,0.059585,59585

238,0.039169,39169

239,0.339841,339841

240,0.457217,457217

241,0.092225,92225

242,0.740289,740289

243,0.468801,468801

244,0.648897,648897

245,0.905665,905665



246,0.487809,487809

247,0.368961,368961

248,0.210177,210177

249,0.275905,275905

250,0.489729,489729

251,0.704321,704321

252,0.453057,453057

253,0.698945,698945

254,0.714049,714049

255,0.218881,218881

256,0.329537,329537

257,0.790785,790785

258,0.088769,88769

259,0.336641,336641

260,0.231617,231617

261,0.687425,687425

262,0.701889,701889

263,0.761601,761601

264,0.791297,791297

265,0.444865,444865

266,0.001409,1409

267,0.077761,77761

268,0.680577,680577

269,0.439105,439105

270,0.995329,995329

271,0.349121,349121

272,0.411457,411457

273,0.766145,766145

274,0.451649,451649

275,0.719681,719681

276,0.135937,135937

277,0.641985,641985



278,0.098369,98369

279,0.013441,13441

280,0.446017,446017

281,0.802625,802625

282,0.823489,823489

283,0.334401,334401

284,0.173697,173697

285,0.904065,904065

286,0.875009,875009

287,0.666561,666561

288,0.190977,190977

289,0.922305,922305

290,0.060929,60929

291,0.473921,473921

292,0.209857,209857

293,0.553345,553345

294,0.949249,949249

295,0.300481,300481

296,0.582337,582337

297,0.613185,613185

298,0.067969,67969

299,0.370241,370241

300,0.100417,100417

301,0.437825,437825

302,0.105089,105089

303,0.187201,187201

304,0.796097,796097

305,0.283265,283265

306,0.108609,108609

307,0.135361,135361

308,0.741377,741377

309,0.725505,725505



310,0.686529,686529

311,0.078721,78721

312,0.848257,848257

313,0.060545,60545

314,0.206849,206849

315,0.961281,961281

316,0.668737,668737

317,0.704385,704385

318,0.997569,997569

319,0.407041,407041

320,0.194817,194817

321,0.593025,593025

322,0.546689,546689

323,0.320001,320001

324,0.658497,658497

325,0.582465,582465

326,0.702209,702209

327,0.484161,484161

328,0.331777,331777

329,0.848705,848705

330,0.872129,872129

331,0.163521,163521

332,0.326657,326657

333,0.287745,287745

334,0.224449,224449

335,0.702081,702081

336,0.395137,395137

337,0.915585,915585

338,0.887169,887169

339,0.123841,123841

340,0.729217,729217

341,0.268225,268225



342,0.148289,148289

343,0.732801,732801

344,0.760897,760897

345,0.801665,801665

346,0.655809,655809

347,0.712961,712961

348,0.962177,962177

349,0.291905,291905

350,0.617729,617729

351,0.728321,728321

352,0.645057,645057

353,0.234945,234945

354,0.002049,2049

355,0.522881,522881

356,0.761537,761537

357,0.246785,246785

358,0.736769,736769

359,0.520641,520641

360,0.703617,703617

361,0.463425,463425

362,0.909889,909889

363,0.425601,425601

364,0.103297,103297

365,0.940865,940865

366,0.969409,969409

367,0.821761,821761

368,0.632577,632577

369,0.055105,55105

370,0.923329,923329

371,0.773121,773121

372,0.803457,803457

373,0.902145,902145



374,0.539649,539649

375,0.423681,423681

376,0.767937,767937

377,0.697985,697985

378,0.546369,546369

379,0.597441,597441

380,0.118017,118017

381,0.178625,178625

382,0.831489,831489

383,0.398401,398401

384,0.685697,685697

385,0.000065,65

386,0.643009,643009

387,0.810561,810561

388,0.342977,342977

389,0.138305,138305

390,0.788929,788929

391,0.297921,297921

392,0.801857,801857

393,0.289345,289345

394,0.837249,837249

395,0.404481,404481

396,0.414337,414337

397,0.269185,269185

398,0.315969,315969

399,0.354241,354241

400,0.972417,972417

401,0.413825,413825

402,0.913089,913089

403,0.651201,651201

404,0.508097,508097

405,0.979265,979265



406,0.676609,676609

407,0.679361,679361

408,0.093377,93377

409,0.541505,541505

410,0.214529,214529

411,0.302721,302721

412,0.640257,640257

413,0.396545,396545

414,0.894849,894849

415,0.465281,465281

416,0.700737,700737

417,0.960385,960385

418,0.045569,45569

419,0.791041,791041

420,0.266817,266817

421,0.169025,169025

422,0.154689,154689

423,0.184001,184001

424,0.570497,570497

425,0.878465,878465

426,0.070209,70209

427,0.428161,428161

428,0.883777,883777

429,0.264705,264705

430,0.200129,200129

431,0.787521,787521

432,0.318657,318657

433,0.223745,223745

434,0.712449,712449

435,0.606081,606081

436,0.627137,627137

437,0.771585,771585



438,0.735169,735169

439,0.907841,907841

440,0.001217,1217

441,0.444225,444225

442,0.556289,556289

443,0.996801,996801

444,0.872897,872897

445,0.697665,697665

446,0.823809,823809

447,0.056961,56961

448,0.714177,714177

449,0.307905,307905

450,0.745729,745729

451,0.752321,752321

452,0.837057,837057

453,0.770945,770945

454,0.290049,290049

455,0.826881,826881

456,0.193537,193537

457,0.702785,702785

458,0.384769,384769

459,0.704641,704641

460,0.175617,175617

461,0.239425,239425

462,0.117889,117889

463,0.089601,89601

464,0.415297,415297

465,0.436865,436865

466,0.937409,937409

467,0.565761,565761

468,0.584577,584577

469,0.671105,671105



470,0.851329,851329

471,0.197121,197121

472,0.195457,195457

473,0.038145,38145

474,0.627649,627649

475,0.127681,127681

476,0.399937,399937

477,0.753985,753985

478,0.994369,994369

479,0.181441,181441

480,0.790017,790017

481,0.554625,554625

482,0.839489,839489

483,0.462401,462401

484,0.197697,197697

485,0.096065,96065

486,0.411009,411009

487,0.954561,954561

488,0.494977,494977

489,0.354305,354305

490,0.516929,516929

491,0.121921,121921

492,0.393857,393857

493,0.025345,25345

494,0.725249,725249

495,0.508481,508481

496,0.246337,246337

497,0.925185,925185

498,0.563969,563969

499,0.338241,338241

500,0.844417,844417

501,0.389825,389825



502,0.721089,721089

503,0.115201,115201

504,0.220097,220097

505,0.675265,675265

506,0.244609,244609

507,0.223361,223361

508,0.445377,445377

509,0.357505,357505

510,0.742529,742529

511,0.526721,526721

512,0.432257,432257

513,0.732545,732545

514,0.582849,582849

515,0.969281,969281

516,0.732737,732737

517,0.216385,216385

518,0.093569,93569

519,0.175041,175041

520,0.338817,338817

521,0.745025,745025

522,0.762689,762689

523,0.048001,48001

524,0.482497,482497

525,0.174465,174465

526,0.438209,438209

527,0.372161,372161

528,0.435777,435777

529,0.680705,680705

530,0.528129,528129

531,0.411521,411521

532,0.310657,310657

533,0.159745,159745



534,0.200449,200449

535,0.510081,510081

536,0.859137,859137

537,0.627585,627585

538,0.583169,583169

539,0.691841,691841

540,0.273217,273217

541,0.620225,620225

542,0.964289,964289

543,0.260801,260801

544,0.984897,984897

545,0.593665,593665

546,0.991809,991809

547,0.400961,400961

548,0.466177,466177

549,0.323905,323905

550,0.873729,873729

551,0.776321,776321

552,0.029057,29057

553,0.306945,306945

554,0.578049,578049

555,0.130881,130881

556,0.625537,625537

557,0.158785,158785

558,0.032769,32769

559,0.888641,888641

560,0.647617,647617

561,0.015425,15425

562,0.325889,325889

563,0.753601,753601

564,0.727297,727297

565,0.932865,932865



566,0.905409,905409

567,0.309761,309761

568,0.536577,536577

569,0.287105,287105

570,0.779329,779329

571,0.621121,621121

572,0.587457,587457

573,0.174145,174145

574,0.715649,715649

575,0.831681,831681

576,0.031937,31937

577,0.809985,809985

578,0.442369,442369

579,0.765441,765441

580,0.462017,462017

581,0.930625,930625

582,0.847489,847489

583,0.526401,526401

584,0.709697,709697

585,0.192065,192065

586,0.179009,179009

587,0.098561,98561

588,0.646977,646977

589,0.570305,570305

590,0.244929,244929

591,0.945921,945921

592,0.985857,985857

593,0.761345,761345

594,0.613249,613249

595,0.612481,612481

596,0.078337,78337

597,0.581185,581185



598,0.811969,811969

599,0.322241,322241

600,0.716417,716417

601,0.365825,365825

602,0.529089,529089

603,0.579201,579201

604,0.932097,932097

605,0.371265,371265

606,0.812609,812609

607,0.767361,767361

608,0.797377,797377

609,0.173505,173505

610,0.270529,270529

611,0.750721,750721

612,0.224257,224257

613,0.068545,68545

614,0.270849,270849

615,0.473281,473281

616,0.764737,764737

617,0.472385,472385

618,0.141569,141569

619,0.559041,559041

620,0.410817,410817

621,0.321025,321025

622,0.370689,370689

623,0.912001,912001

624,0.394497,394497

625,0.470465,470465

626,0.806209,806209

627,0.316161,316161

628,0.987777,987777

629,0.096705,96705



630,0.856129,856129

631,0.035521,35521

632,0.302657,302657

633,0.095745,95745

634,0.688449,688449

635,0.414081,414081

636,0.091137,91137

637,0.483585,483585

638,0.431169,431169

639,0.475841,475841

640,0.545217,545217

641,0.796225,796225

642,0.372289,372289

643,0.524801,524801

644,0.096897,96897

645,0.489665,489665

646,0.159809,159809

647,0.744961,744961

648,0.218177,218177

649,0.339905,339905

650,0.001729,1729

651,0.800321,800321

652,0.221057,221057

653,0.842945,842945

654,0.866049,866049

655,0.434881,434881

656,0.057537,57537

657,0.614785,614785

658,0.680769,680769

659,0.072641,72641

660,0.119617,119617

661,0.791425,791425



662,0.533889,533889

663,0.417601,417601

664,0.039297,39297

665,0.428865,428865

666,0.873409,873409

667,0.053761,53761

668,0.488577,488577

669,0.903105,903105

670,0.707329,707329

671,0.045121,45121

672,0.979457,979457

673,0.310145,310145

674,0.803649,803649

675,0.535681,535681

676,0.663937,663937

677,0.865985,865985

678,0.890369,890369

679,0.349441,349441

680,0.134017,134017

681,0.306625,306625

682,0.855489,855489

683,0.590401,590401

684,0.221697,221697

685,0.288065,288065

686,0.947009,947009

687,0.242561,242561

688,0.798977,798977

689,0.786305,786305

690,0.972929,972929

691,0.769921,769921

692,0.577857,577857

693,0.497345,497345



694,0.501249,501249

695,0.716481,716481

696,0.910337,910337

697,0.237185,237185

698,0.059969,59969

699,0.306241,306241

700,0.588417,588417

701,0.341825,341825

702,0.337089,337089

703,0.043201,43201

704,0.644097,644097

705,0.067265,67265

706,0.380609,380609

707,0.311361,311361

708,0.149377,149377

709,0.989505,989505

710,0.798529,798529

711,0.974721,974721

712,0.016257,16257

713,0.404545,404545

714,0.958849,958849

715,0.977281,977281

716,0.796737,796737

717,0.728385,728385

718,0.189569,189569

719,0.943041,943041

720,0.482817,482817

721,0.897025,897025

722,0.978689,978689

723,0.776001,776001

724,0.306497,306497

725,0.766465,766465



726,0.174209,174209

727,0.260161,260161

728,0.539777,539777

729,0.512705,512705

730,0.184129,184129

731,0.659521,659521

732,0.294657,294657

733,0.031745,31745

734,0.176449,176449

735,0.318081,318081

736,0.323137,323137

737,0.339585,339585

738,0.279169,279169

739,0.259841,259841

740,0.817217,817217

741,0.972225,972225

742,0.780289,780289

743,0.788801,788801

744,0.208897,208897

745,0.385665,385665

746,0.327809,327809

747,0.088961,88961

748,0.970177,970177

749,0.355905,355905

750,0.129729,129729

751,0.824321,824321

752,0.413057,413057

753,0.378945,378945

754,0.154049,154049

755,0.738881,738881

756,0.489537,489537

757,0.070785,70785



758,0.328769,328769

759,0.256641,256641

760,0.591617,591617

761,0.567425,567425

762,0.741889,741889

763,0.081601,81601

764,0.351297,351297

765,0.924865,924865

766,0.841409,841409

767,0.797761,797761

768,0.440577,440577

769,0.519105,519105

770,0.635329,635329

771,0.469121,469121

772,0.371457,371457

773,0.446145,446145

774,0.891649,891649

775,0.239681,239681

776,0.295937,295937

777,0.921985,921985

778,0.338369,338369

779,0.933441,933441

780,0.806017,806017

781,0.682625,682625

782,0.863489,863489

783,0.654401,654401

784,0.733697,733697

785,0.384065,384065

786,0.715009,715009

787,0.386561,386561

788,0.950977,950977

789,0.002305,2305



790,0.700929,700929

791,0.593921,593921

792,0.169857,169857

793,0.233345,233345

794,0.389249,389249

795,0.820481,820481

796,0.742337,742337

797,0.893185,893185

798,0.307969,307969

799,0.290241,290241

800,0.460417,460417

801,0.317825,317825

802,0.145089,145089

803,0.507201,507201

804,0.356097,356097

805,0.763265,763265

806,0.948609,948609

807,0.855361,855361

808,0.501377,501377

809,0.805505,805505

810,0.326529,326529

811,0.198721,198721

812,0.808257,808257

813,0.740545,740545

814,0.646849,646849

815,0.481281,481281

816,0.828737,828737

817,0.984385,984385

818,0.237569,237569

819,0.327041,327041

820,0.554817,554817

821,0.473025,473025



822,0.586689,586689

823,0.640001,640001

824,0.218497,218497

825,0.062465,62465

826,0.542209,542209

827,0.204161,204161

828,0.091777,91777

829,0.928705,928705

830,0.512129,512129

831,0.283521,283521

832,0.286657,286657

833,0.967745,967745

834,0.664449,664449

835,0.222081,222081

836,0.555137,555137

837,0.195585,195585

838,0.127169,127169

839,0.043841,43841

840,0.089217,89217

841,0.148225,148225

842,0.188289,188289

843,0.052801,52801

844,0.320897,320897

845,0.281665,281665

846,0.495809,495809

847,0.432961,432961

848,0.722177,722177

849,0.371905,371905

850,0.257729,257729

851,0.848321,848321

852,0.605057,605057

853,0.914945,914945



854,0.442049,442049

855,0.042881,42881

856,0.921537,921537

857,0.526785,526785

858,0.976769,976769

859,0.440641,440641

860,0.063617,63617

861,0.343425,343425

862,0.949889,949889

863,0.745601,745601

864,0.663297,663297

865,0.420865,420865

866,0.809409,809409

867,0.541761,541761

868,0.392577,392577

869,0.135105,135105

870,0.563329,563329

871,0.893121,893121

872,0.763457,763457

873,0.582145,582145

874,0.979649,979649

875,0.943681,943681

876,0.927937,927937

877,0.977985,977985

878,0.786369,786369

879,0.517441,517441

880,0.478017,478017

881,0.058625,58625

882,0.871489,871489

883,0.718401,718401

884,0.245697,245697

885,0.480065,480065



886,0.483009,483009

887,0.530561,530561

888,0.102977,102977

889,0.218305,218305

890,0.428929,428929

891,0.417921,417921

892,0.761857,761857

893,0.969345,969345

894,0.277249,277249

895,0.924481,924481

896,0.574337,574337

897,0.549185,549185

898,0.555969,555969

899,0.274241,274241

900,0.332417,332417

901,0.293825,293825

902,0.953089,953089

903,0.971201,971201

904,0.068097,68097

905,0.459265,459265

906,0.516609,516609

907,0.399361,399361

908,0.853377,853377

909,0.621505,621505

910,0.854529,854529

911,0.422721,422721

912,0.600257,600257

913,0.076545,76545

914,0.334849,334849

915,0.985281,985281

916,0.860737,860737

917,0.240385,240385



918,0.285569,285569

919,0.711041,711041

920,0.626817,626817

921,0.049025,49025

922,0.194689,194689

923,0.504001,504001

924,0.130497,130497

925,0.358465,358465

926,0.910209,910209

927,0.148161,148161

928,0.643777,643777

929,0.344705,344705

930,0.840129,840129

931,0.907521,907521

932,0.278657,278657

933,0.903745,903745

934,0.152449,152449

935,0.126081,126081

936,0.787137,787137

937,0.051585,51585

938,0.975169,975169

939,0.827841,827841

940,0.361217,361217

941,0.324225,324225

942,0.596289,596289

943,0.316801,316801

944,0.432897,432897

945,0.177665,177665

946,0.663809,663809

947,0.776961,776961

948,0.474177,474177

949,0.387905,387905



950,0.385729,385729

951,0.872321,872321

952,0.797057,797057

953,0.450945,450945

954,0.730049,730049

955,0.346881,346881

956,0.353537,353537

957,0.982785,982785

958,0.624769,624769

959,0.624641,624641

960,0.535617,535617

961,0.119425,119425

962,0.157889,157889

963,0.409601,409601

964,0.975297,975297

965,0.916865,916865

966,0.777409,777409

967,0.285761,285761

968,0.344577,344577

969,0.751105,751105

970,0.491329,491329

971,0.317121,317121

972,0.155457,155457

973,0.718145,718145

974,0.067649,67649

975,0.647681,647681

976,0.559937,559937

977,0.033985,33985

978,0.234369,234369

979,0.101441,101441

980,0.150017,150017

981,0.434625,434625



982,0.879489,879489

983,0.782401,782401

984,0.757697,757697

985,0.576065,576065

986,0.251009,251009

987,0.674561,674561

988,0.254977,254977

989,0.434305,434305

990,0.156929,156929

991,0.241921,241921

992,0.353857,353857

993,0.705345,705345

994,0.165249,165249

995,0.028481,28481

996,0.406337,406337

997,0.205185,205185

998,0.803969,803969

999,0.258241,258241

1000,0.204417,204417